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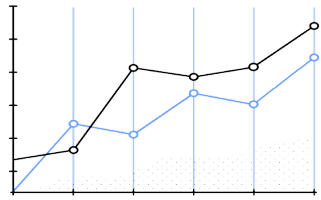
PROJECT ON

ENERGY CONSUMPTION ANALYSIS

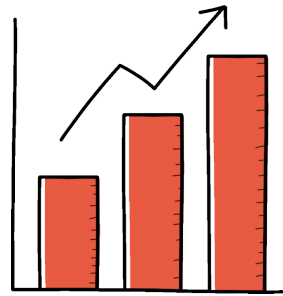
Presented by
Bhargav Kunapareddy
Tummala Lahari



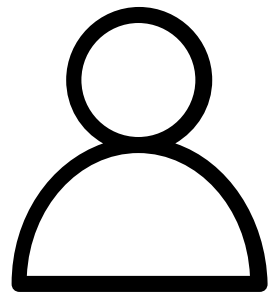
OBJECTIVES



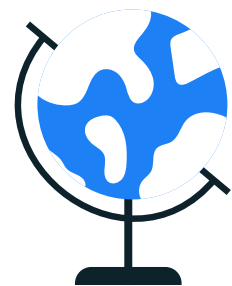
→ **General & Comparative Analysis**



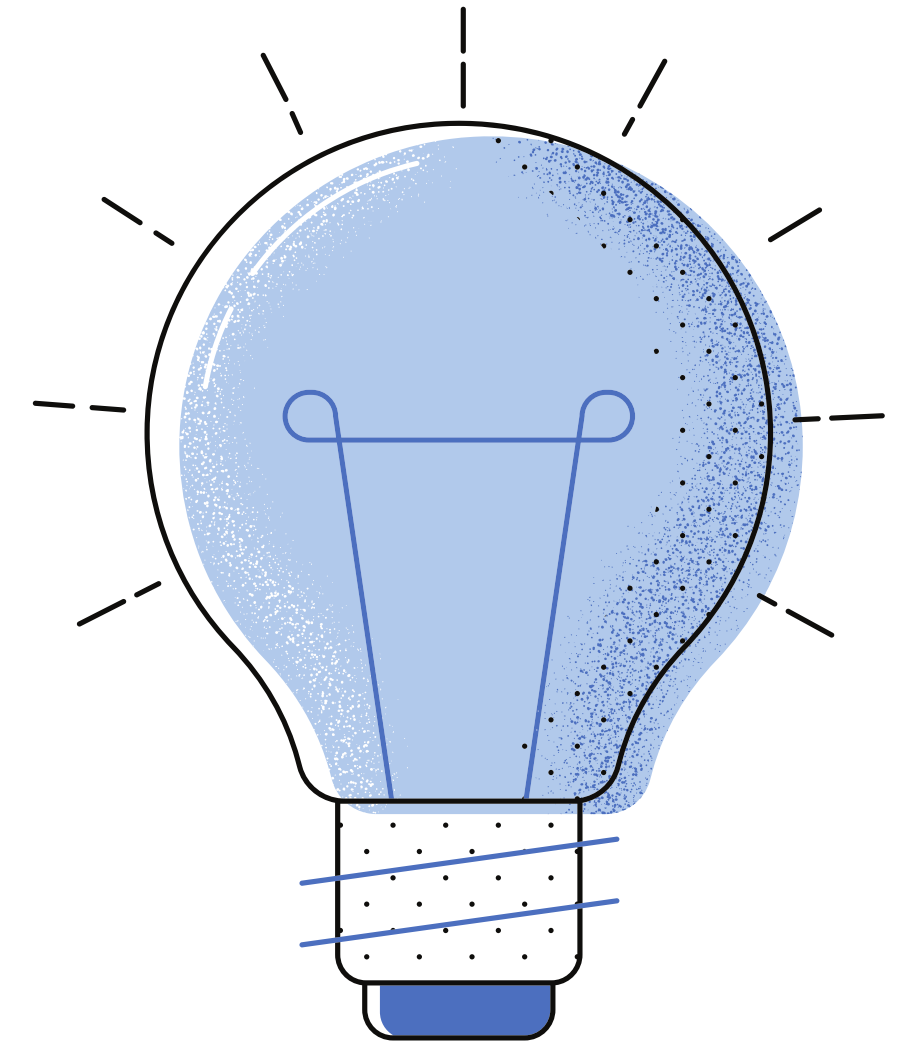
→ **Trend Analysis Over Time**



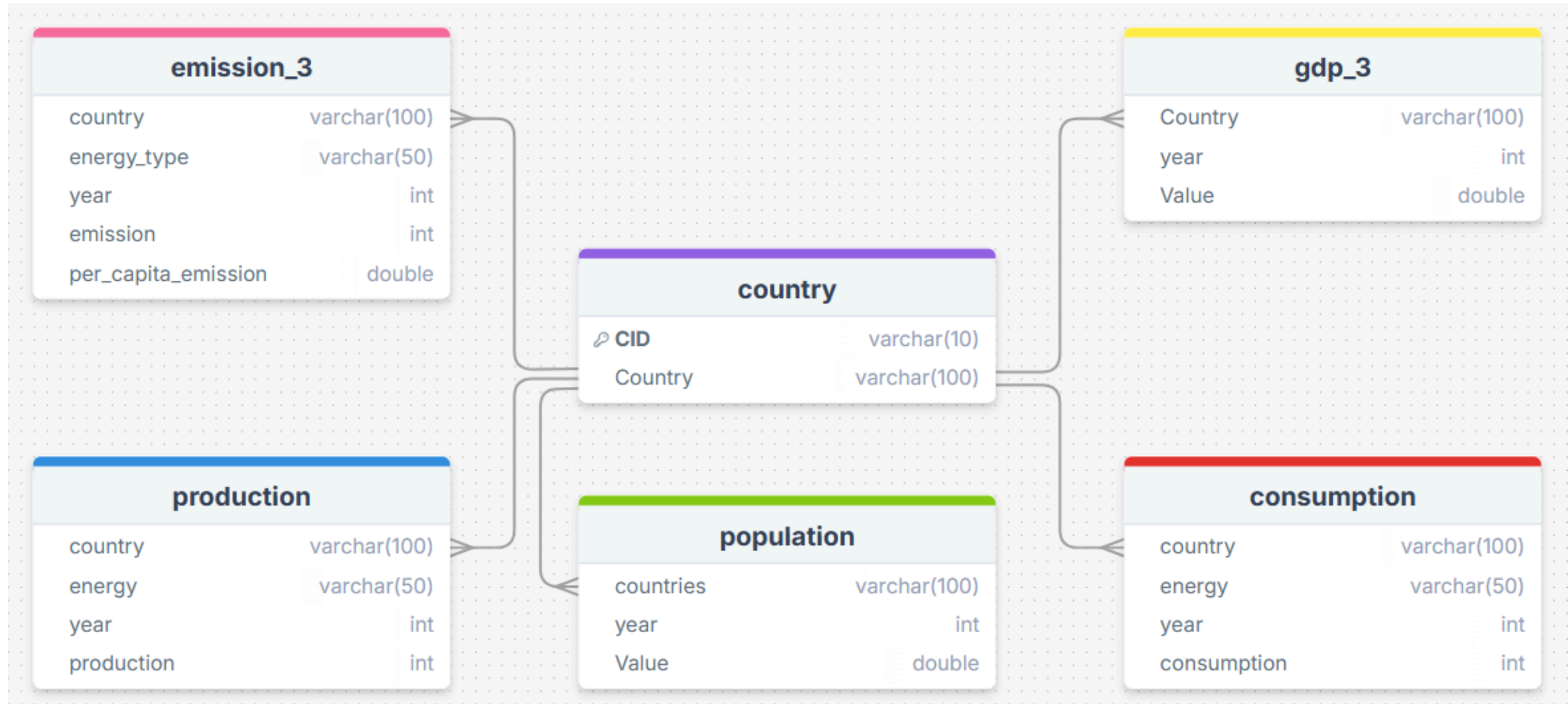
→ **Ratio & Per Capita Analysis**



→ **Global Comparisons**



ER DIAGRAM



SCHEMA

Table	Columns	Relationships	Number of records
Country	Country, CID (PK)	1-to-M with all tables	231
production	Country (FK), energy, year, production	M-to-1 Country	5297
population	Countries (FK), year, Value	M-to-1 Country	1001
consumption	Country (FK), energy, year, consumption	M-to-1 Country	5281
emission_3	Country (FK), energy_type, year, emission, per_capita_emission	M-to-1 Country	3517
gdp_3	Country (FK), year, Value	M-to-1 Country	1000

DATA PREPARATION – VIEWS

Energy types in emission table

- **CO2 emissions (MMtonnes CO2)**
- Consumed natural gas (MMtonnes CO2)
- Coal and coke (MMtonnes CO2)
- Petroleum and other liquids (MMtonnes CO2)



energy_type	year	emission
Petroleum and other liquids (MMtonnes CO2)	2020	7.2639375
CO2 emissions (MMtonnes CO2)	2020	9.366235986
Consumed natural gas (MMtonnes CO2)	2020	0.166604406
Coal and coke (MMtonnes CO2)	2020	1.93569408

```
CREATE VIEW EMISSION_3 AS
```

```
SELECT * FROM EMISSION WHERE ENERGY_TYPE <> "CO2 emissions (MMtonnes CO2)";
```

```
CREATE VIEW EMISSION_TOTAL AS
```

```
SELECT * FROM EMISSION WHERE ENERGY_TYPE = 'CO2 emissions (MMtonnes CO2)';
```

DATA PREPARATION – VIEWS

Energy types in production and consumption tables

- Nuclear (quad Btu)
- Natural gas (quad Btu)
- Coal (quad Btu)
- **Nuclear, renewables, and other (quad Btu)**
- Renewables and other (quad Btu)
- Petroleum and other liquids (quad Btu)

year	energy	production
2020	Nuclear, renewables, and other (quad Btu)	0.0035963970
2020	Nuclear (quad Btu)	0.0000000000
2020	Petroleum and other liquids (quad Btu)	0.0000000000
2020	Renewables and other (quad Btu)	0.0035963970
2020	Natural gas (quad Btu)	0.0031401220
2020	Coal (quad Btu)	0.0353306920



```
CREATE VIEW PRODUCTION AS
```

```
SELECT * FROM PRODUCTION_V WHERE ENERGY <> 'Nuclear, renewables, and other (quad Btu)';
```

```
CREATE VIEW CONSUMPTION AS
```

```
SELECT * FROM CONSUMPTION_V WHERE ENERGY <> 'Nuclear, renewables, and other (quad Btu)';
```

General & Comparative Analysis



1. What is the total emission per country for the most recent year available?

Approach	Concepts Used
Identified the most recent year available for each country using MAX(YEAR). Used a subquery to filter only latest year records. Selected total_emissions for each country, Ordered the result to find the top 5 emitters.	• Aggregate function: (MAX(YEAR)) • GROUP BY • Subquery • Multi-column filtering (country, year) • ORDER BY - Sorting • LIMIT

```

SELECT COUNTRY, YEAR, ROUND(EMISSION,2) AS TOTAL_EMISSION
FROM EMISSION_TOTAL
WHERE (COUNTRY, YEAR) IN (
    SELECT COUNTRY, MAX(YEAR)
    FROM EMISSION_3
    GROUP BY COUNTRY
)
ORDER BY TOTAL_EMISSION DESC
LIMIT 5;

```

country	year	TOTAL_EMISSION
China	2023	12195.66
United States	2023	4795.37
India	2023	2820.66
Russia	2023	1844.49
Japan	2023	960.23

In 2023, China has the highest total emissions among the countries analyzed. It produces much more emissions compared to the United States and India. The gap between China and the second highest country is very large.

2. What are the top 5 countries by GDP in the most recent year?

Approach	Concepts Used
Identified the most recent year using MAX(YEAR). Used a scalar subquery to filter records for that year. Ranked countries by GDP in descending order. Selected the top 5 countries using LIMIT.	• Subquery (Scalar subquery) • Filtering (WHERE CLAUSE) • Sorting (ORDER BY) • Limiting Results (LIMIT)


```
SELECT COUNTRY, YEAR , VALUE AS GDP
FROM GDP_3
WHERE YEAR = (SELECT MAX(YEAR) FROM GDP_3)
ORDER BY VALUE DESC
LIMIT 5;
```

COUNTRY	YEAR	GDP
China	2024	28673.24
United States	2024	22679.47
India	2024	11660.21
Japan	2024	5179.704
Germany	2024	4463.949

The data shows that China and the United States have much higher GDP compared to the other countries in the list. India is growing strongly and holds the third position, while Japan and Germany follow with comparatively lower GDP values.

3. Compare energy production and consumption by country and year.

Approach	Concepts Used
Joined the production and consumption tables on country, year, and energy type. Summed production and consumption per country per year. Calculated net difference: SUM(production) - SUM(consumption) Used a CASE statement to classify SURPLUS DEFICIT BALANCED	• INNER JOIN • SUM() for totals • GROUP BY for aggregation • CASE WHEN for conditional classification • ORDER BY for sorting.

```

SELECT P.COUNTRY,
       P.YEAR,
       SUM(P.PRODUCTION) AS TOTAL_PRODUCTION,
       SUM(C.CONSUMPTION) AS TOTAL_CONSUMPTION,
       SUM(P.PRODUCTION) - SUM(C.CONSUMPTION) AS NET_COMPARISON,
       CASE
           WHEN SUM(P.PRODUCTION) > SUM(C.CONSUMPTION) THEN "SURPLUS"
           WHEN SUM(P.PRODUCTION) < SUM(C.CONSUMPTION) THEN "DEFICIT"
           ELSE "BALANCED"
       END AS ENERGY_STATUS
FROM PRODUCTION P
JOIN CONSUMPTION C
ON P.COUNTRY = C.COUNTRY
AND P.YEAR = C.YEAR
AND P.ENERGY = C.ENERGY
GROUP BY P.COUNTRY, P.YEAR
ORDER BY P.COUNTRY, P.YEAR;

```

country	year	TOTAL_PRODUCTION	TOTAL_CONSUMPTION	NET_COMPARISON	ENERGY_STATUS
Afghanistan	2020	0.0420672110	0.1600695110	-0.1180023000	DEFICIT
Afghanistan	2021	0.0571382690	0.1623132150	-0.1051749460	DEFICIT
Afghanistan	2022	0.0962556700	0.1539091990	-0.0576535290	DEFICIT
Afghanistan	2023	0.0180285970	0.1401014500	-0.1220728530	DEFICIT
Albania	2020	0.0611128690	0.0720539720	-0.0109411030	DEFICIT
Albania	2021	0.0718560580	0.0869745330	-0.0151184750	DEFICIT
Albania	2022	0.0664429490	0.0882542450	-0.0218112960	DEFICIT
Albania	2023	0.0721412460	0.0770579320	-0.0049166860	DEFICIT
Algeria	2020	5.9694254450	2.4708052620	3.4986201830	SURPLUS
Algeria	2021	6.7318767720	2.6509060540	4.0809707180	SURPLUS
Algeria	2022	6.7245775530	2.7671857430	3.9573918100	SURPLUS

Net difference trends highlight which countries are energy exporters versus import-dependent.

ENERGY_STATUS	TOTAL_COUNT	PERCENTAGE
DEFICIT	682	76.46
SURPLUS	182	20.40
BALANCED	28	3.14

Country-year combinations

4. Which energy types contribute most to emissions across all countries?

Approach	Concepts Used
Summed total emissions per energy type across all countries using SUM(). Grouped results by ENERGY_TYPE to get totals per type. Ordered results in descending order to see which energy types contributes most.	• SUM() • GROUP BY • ORDER BY • ROUND()

```
SELECT ENERGY_TYPE, ROUND(SUM(EMISSION), 2) AS TOTAL_EMISSION
FROM EMISSION_3
GROUP BY ENERGY_TYPE
ORDER BY TOTAL_EMISSION DESC;
```

energy_type	TOTAL_EMISSION
Coal and coke (MMtonnes CO2)	63959.19
Petroleum and other liquids (MMtonnes CO2)	47305.91
Consumed natural gas (MMtonnes CO2)	31466.51

Fossil fuels (coal, petroleum, natural gas) are the largest contributors to global emissions.

Reducing their use and investing in clean energy can have the biggest impact on lowering carbon emissions.

Trend Analysis over Time



5. How have global emissions changed year over year?

Approach	Concepts Used
Used a CTE (WITH clause) to calculate total emissions per year. Calculated year-over-year change using LAG() window function. Classified the change as INCREASED, DECREASED, or Starting year.	• CTE (Common Table Expression) • Aggregate function: SUM() • Window function: LAG() • CASE statements for categorization • GROUP BY • ORDER BY

```
WITH YEAR_DATA AS(
SELECT YEAR, SUM(EMISSION) AS TOTAL_EMISSION,
SUM(EMISSION) - LAG(SUM(EMISSION)) OVER (ORDER BY YEAR) AS Y_T_Y_CHANGE
FROM EMISSION_TOTAL
GROUP BY YEAR
)
SELECT YEAR, TOTAL_EMISSION, Y_T_Y_CHANGE,
CASE
WHEN Y_T_Y_CHANGE IS NULL THEN "Starting year"
WHEN Y_T_Y_CHANGE > 0 THEN "INCREASED"
WHEN Y_T_Y_CHANGE < 0 THEN "DECREASED"
ELSE "NO CHANGE"
END AS GLOBAL_CHANGE
FROM YEAR_DATA
ORDER BY YEAR;
```

Global emissions have consistently increased from 2020 to 2023. Shows a growing demand for energy and emphasis on clean energy adoption to curb rising emissions.

YEAR	TOTAL_EMISSION	Y_T_Y_CHANGE	GLOBAL_CHANGE
2020	33927.610924815985	NULL	Starting year
2021	35495.71408284799	1568.103158032005	INCREASED
2022	36229.097987573994	733.3839047260044	INCREASED
2023	37079.18123464902	850.0832470750247	INCREASED

6. What is the trend in GDP for each country over the given years?

Approach	Concepts Used
Used a CTE (WITH clause) to prepare GDP data per country per year. Applied the LAG() window function to get the previous year's GDP. Calculated yearly difference (VALUE - PREV_GDP) and used CASE to indicate Increase, Decrease, or No change.	• CTE (Common Table Expression) • Window function: LAG() • Aggregate per row calculation (VALUE - PREV_GDP) • CASE statements for trend classification • ORDER BY for chronological display

```
WITH GDP_DATA AS(
SELECT COUNTRY, YEAR, VALUE,
LAG(VALUE) OVER (PARTITION BY COUNTRY ORDER BY YEAR) AS PREV_GDP
FROM GDP_3)
```

```
SELECT COUNTRY, YEAR, VALUE AS GDP,
ROUND(VALUE - PREV_GDP,4) AS DIFFERENCE,
CASE
WHEN VALUE > PREV_GDP
THEN "Increased"
WHEN VALUE < PREV_GDP
THEN "Decreased"
ELSE "No change"
END AS DIRECTION
FROM GDP_DATA
ORDER BY COUNTRY, YEAR;
```

COUNTRY	YEAR	GDP	DIFFERENCE	DIRECTION
Afghanistan	2020	83.21645	HULL	No change
Afghanistan	2021	65.95827	-17.2582	Decreased
Afghanistan	2022	61.84236	-4.1159	Decreased
Afghanistan	2023	58.90645	-2.9359	Decreased
Afghanistan	2024	57.83112	-1.0753	Decreased
Albania	2020	36.78752	HULL	No change
Albania	2021	40.04762	3.2601	Increased
Albania	2022	41.99279	1.9452	Increased
Albania	2023	43.63172	1.6389	Increased
Albania	2024	45.2091	1.5774	Increased

GDP trends vary by country: Some countries show a continuous decrease over the years, others show a steady increase.

Helps identify growing vs struggling economies, which can guide energy policy and investment decisions.

7. How has population growth affected total emissions in each country?

Approach	Concepts Used
Joined EMISSION_TOTAL with POPULATION on COUNTRY and YEAR. Used LAG() to calculate year-on-year changes in emissions and population. Labeled trends with CASE. Ordered results by COUNTRY and YEAR	• CTE (WITH clause) for intermediate calculations • Window Function: LAG() to get previous year values • CASE Statement for trend classification • JOIN between tables (EMISSION_TOTAL & POPULATION) • Ordering and sorting

```
WITH EMISSION_POP_TREND AS(
SELECT E.COUNTRY, E.YEAR, E.EMISSION, P.VALUE AS POPULATION,
(E.EMISSION - LAG(E.EMISSION) OVER (PARTITION BY E.COUNTRY ORDER BY E.YEAR)) AS EMI_CHANGE,
(P.VALUE - LAG(P.VALUE) OVER (PARTITION BY P.COUNTRIES ORDER BY P.YEAR)) AS POP_CHANGE
FROM EMISSION_TOTAL E
JOIN POPULATION P
ON E.COUNTRY = P.COUNTRIES AND
E.YEAR = P.YEAR)
SELECT COUNTRY, YEAR, EMISSION, POPULATION, EMI_CHANGE,POP_CHANGE,
CASE
WHEN EMI_CHANGE IS NULL THEN "START_YEAR"
WHEN EMI_CHANGE > 0 THEN "Emission increased"
WHEN EMI_CHANGE < 0 THEN "Emission decreased"
ELSE "NO CHANGE"
END AS EMISSION_TREND,
CASE
WHEN POP_CHANGE IS NULL THEN "START_YEAR"
WHEN POP_CHANGE > 0 THEN "Population increased"
WHEN POP_CHANGE < 0 THEN "Population decreased"
ELSE "NO CHANGE"
END AS POPULATION_TREND
FROM EMISSION_POP_TREND
ORDER BY COUNTRY, YEAR;
```

Higher population growth contributes to increased energy demand and emissions, but efficiency and energy mix also influence total impact.

COUNTRY	YEAR	EMISSION	POPULATION	EMI_CHANGE	POP_CHANGE	EMISSION_TREND	POPULATION_TREND
Afghanistan	2020	9.366235986	39068.98	NULL	NULL	START_YEAR	START_YEAR
Afghanistan	2021	9.936315561	40000.41	0.5700795750000012	931.43000000000003	Emission increased	Population increased
Afghanistan	2022	9.302426307	40578.84	-0.6338892540000014	578.4299999999993	Emission decreased	Population increased
Afghanistan	2023	7.756799675	41454.76	-1.5456266319999994	875.92000000000055	Emission decreased	Population increased
Albania	2020	3.309498526	2871.954	NULL	NULL	START_YEAR	START_YEAR
Albania	2021	4.002473171	2849.635	0.6929746450000001	-22.318999999999996	Emission increased	Population decreased
Albania	2022	4.294648618	2827.608	0.292175447	-22.0270000000000044	Emission increased	Population decreased
Albania	2023	3.392413045	2811.655	-0.902235573	-15.9529999999999975	Emission decreased	Population decreased

8. Has energy consumption increased or decreased over the years for major economies?

Approach	Concepts Used
Identified top economies by GDP. Calculated total energy consumption per year per country. Used LAG() to find previous year's consumption and compute year-on-year change. Categorized consumption trend using CASE Filtered results for major economies.	• Common Table Expressions (CTEs) • Aggregate functions (SUM()) • Window functions (LAG()) • CASE for conditional labeling • Filtering with subquery • Year-on-year change calculation

```
WITH MAJ_ECO AS(
SELECT COUNTRY, SUM(VALUE) AS TOTAL_GDP FROM GDP_3 GROUP BY COUNTRY ORDER BY TOTAL_GDP DESC LIMIT 2
),
TOT_CON AS(
SELECT COUNTRY, YEAR, SUM(CONSUMPTION) AS TOTAL_CONSUMPTION
FROM CONSUMPTION group by COUNTRY, YEAR),
CT AS(
SELECT COUNTRY, YEAR, TOTAL_CONSUMPTION,
LAG(TOTAL_CONSUMPTION) OVER (PARTITION BY COUNTRY ORDER BY YEAR) AS PREV_CONS
FROM TOT_CON
)
SELECT CT.COUNTRY, CT.YEAR, CT.TOTAL_CONSUMPTION,
CT.TOTAL_CONSUMPTION - CT.PREV_CONS AS CONSUMPTION_CHANGE ,
CASE
WHEN PREV_CONS IS NULL THEN "First Year"
WHEN TOTAL_CONSUMPTION > PREV_CONS THEN "Increased"
WHEN TOTAL_CONSUMPTION < PREV_CONS THEN "Decreased"
ELSE "No change"
END AS CON_TREND
FROM CT WHERE CT.COUNTRY IN (SELECT COUNTRY FROM MAJ_ECO)
ORDER BY COUNTRY, YEAR;
```

COUNTRY	YEAR	TOTAL_CONSUMPTION	CONSUMPTION_CHANGE	CON_TREND
China	2020	144.4419644600	NULL	First Year
China	2021	151.6368096650	7.1948452050	Increased
China	2022	153.5197192800	1.8829096150	Increased
China	2023	161.8967937040	8.3770744240	Increased
United States	2020	90.0365525220	NULL	First Year
United States	2021	94.6659497760	4.6293972540	Increased
United States	2022	96.1249983110	1.4590485350	Increased
United States	2023	95.1090718050	-1.0159265060	Decreased

Energy consumption is rising in most major economies, but the US shows a slight drop in 2023, indicating potential efficiency gains or structural changes in energy use.

9. What is the average yearly change in emissions per capita for each country?

Approach	Concepts Used
Calculated per capita emissions for each country (EMISSION / POPULATION). Used LAG() to find year-on-year changes in per capita emissions. Averaged these yearly changes per country to get the average yearly change.	• Window Functions (LAG()) • Per capita calculation • Aggregation (AVG()) • JOIN tables on country and year

```

WITH A AS(
SELECT E.COUNTRY, E.YEAR,
(E.EMISSION/P.VALUE) - LAG(E.EMISSION/P.VALUE) OVER (PARTITION BY E.COUNTRY ORDER BY E.YEAR)
AS YOY_CHANGE_PCE FROM EMISSION_TOTAL E
JOIN POPULATION P
ON E.COUNTRY = P.COUNTRIES AND
E.YEAR = P.YEAR
)
SELECT COUNTRY, AVG(YOY_CHANGE_PCE) AS AVG_YOY_PCE
FROM A
GROUP BY COUNTRY
ORDER BY AVG_YOY_PCE DESC;

```

Small increases in per capita emissions indicate improved energy efficiency or shifts toward cleaner energy sources.

Large positive changes highlight countries where emission reduction policies are needed most.

COUNTRY	AVG_YOY_PCE
Netherlands Antilles	0.0010890091157186796
Bosnia and Herzegovina	0.0009433193248508465
Qatar	0.0008073452074844565
Kuwait	0.0007913275434875076
Singapore	0.00069792525026309
Bahrain	0.0006614396223203062
U.S. Virgin Islands	0.0005908375479073714
Oman	0.0005863660411002964
Russia	0.0004635765134098045
Iran	0.00045796489989445376

Ratio & Per Capita Analysis



10. What is the emission-to-GDP ratio for each country by year?

Approach	Concepts Used
Joined EMISSION_TOTAL with GDP_3 on COUNTRY and YEAR. Calculated the emission-to-GDP ratio by dividing emissions by GDP for each country-year. Ordered results by country and year to observe trends over time.	• JOIN – combining emissions and GDP • Arithmetic calculation – emission / GDP. • ORDER BY – to sort results for readability.

```

SELECT
    E.COUNTRY,
    E.YEAR,
    ROUND(E.EMISSION/G.VALUE, 4) AS EMISSION_GDP_RATIO
FROM EMISSION_TOTAL E
JOIN GDP_3 G
    ON E.COUNTRY = G.COUNTRY
    AND E.YEAR = G.YEAR
ORDER BY E.COUNTRY, E.YEAR;

```

High ratio → carbon-intensive economy
 Low ratio → more efficient growth

country	year	EMISSION_GDP_RATIO
Afghanistan	2020	0.1126
Afghanistan	2021	0.1506
Afghanistan	2022	0.1504
Afghanistan	2023	0.1317
Albania	2020	0.09
Albania	2021	0.0999
Albania	2022	0.1023
Albania	2023	0.0778
Algeria	2020	0.2662
Algeria	2021	0.2753
Algeria	2022	0.2768
Algeria	2023	0.2746

11. What is the energy consumption per capita for each country over the last decade?

Approach	Concepts Used
Calculated total energy consumption per country per year. Joined with population to get per capita consumption. Rounded values for clarity. Sorted by country and year.	• Subquery for aggregation • JOIN • Division for per capita • ROUND function • ORDER BY

```

SELECT
    C.COUNTRY,
    C.YEAR,
    ROUND(C.TOTAL_CONSUMPTION/P.VALUE, 8) AS PER_CAPITA_CONS
FROM (
    SELECT
        COUNTRY,
        YEAR,
        SUM(CONSUMPTION) AS TOTAL_CONSUMPTION
    FROM CONSUMPTION
    GROUP BY COUNTRY, YEAR
) C
JOIN POPULATION P
    ON C.COUNTRY = P.COUNTRIES
    AND C.YEAR = P.YEAR
ORDER BY C.COUNTRY, C.YEAR;

```

COUNTRY	YEAR	PER_CAPITA_CONS
Afghanist...	2020	0.00000041
Afghanist...	2021	0.000000406
Afghanist...	2022	0.000000379
Afghanist...	2023	0.000000338
Albania	2020	0.000002509
Albania	2021	0.000003052
Albania	2022	0.000003121
Albania	2023	0.000002741

Countries with rising per capita consumption may need energy efficiency policies.
Some countries show decreasing per capita consumption, suggesting improved efficiency or cleaner energy use.

12. How does energy production per capita vary across countries?

Approach	Concepts Used
Aggregated total energy production by country and year. Calculated production per capita Averaged per-capita production for each country over the available years . Sorted results to show countries with the highest average production per person.	• Subquery for total production (SUM(PRODUCTION)) • Join with population data • Per capita calculation • Aggregation with AVG() • Ordering results


```

WITH PRO AS(
SELECT COUNTRY, YEAR, SUM(PRODUCTION) AS TOTAL_PRODUCTION FROM PRODUCTION GROUP BY COUNTRY,year
),
PER_CAPITA AS(
SELECT P.COUNTRY, P.YEAR, (P.TOTAL_PRODUCTION/PO.VALUE) AS PER_CAPITA_PROD FROM PRO P
JOIN POPULATION PO ON
P.COUNTRY = PO.COUNTRIES AND
P.YEAR = PO.YEAR
)

SELECT COUNTRY, AVG(PER_CAPITA_PROD)
AS AVG_PROD_PER_CAPITA FROM PER_CAPITA
GROUP BY COUNTRY
ORDER BY AVG_PROD_PER_CAPITA DESC LIMIT 10;

```

Based on the top 10 countries by average per-capita energy production, smaller or energy-rich countries (e.g., Qatar, Norway, Kuwait) have the highest per-capita production, while larger-population countries (e.g., China, South Africa) have lower per-capita production despite high total output.

COUNTRY	AVG_PROD_PER_CAPITA
Qatar	0.0034268054258363013
Norway	0.0016560802026935203
Kuwait	0.001463331552026933
Brunei	0.0013738227326009758
United Arab Emirates	0.0010276815491238498
Saudi Arabia	0.0008637890028690346
Trinidad and Tobago	0.0007799016565626575
Oman	0.0007605060651646072
Bahrain	0.0007089619454267149
Australia	0.0006829348472523485

13. Which countries have the highest energy consumption relative to GDP?

Approach	Concepts Used
Summed energy consumption per country/year. Joined with GDP to calculate energy-to-GDP ratio. Averaged over years and selected top 10 countries.	• SUM() • JOIN • GROUP BY • ORDER BY • LIMIT

```
WITH CON AS(
SELECT COUNTRY, YEAR, SUM(CONSUMPTION) AS TOTAL_CONS FROM CONSUMPTION GROUP BY COUNTRY , YEAR
),
```

```
RATIO AS(
SELECT C.COUNTRY, C.YEAR, (C.TOTAL_CONS/G.VALUE) AS RATIO FROM CON C
JOIN GDP_3 G ON
C.COUNTRY = G.COUNTRY AND
C.YEAR = G.YEAR
)
```

```
SELECT COUNTRY, AVG(RATIO) AS AVG_ENERGY_RATIO FROM RATIO
GROUP BY COUNTRY
ORDER BY AVG_ENERGY_RATIO DESC LIMIT 10;
```

COUNTRY	AVG_ENERGY_RATIO
Netherlands Antilles	0.0577154659341407
Trinidad and Tobago	0.019548547295605123
Turkmenistan	0.01868818011349008
North Korea	0.017703335754069658
U.S. Virgin Islands	0.01749392858896457
Libya	0.010681151584129763
Bahrain	0.010680062437356716
Iran	0.010220266310976701
Venezuela	0.009788649573785176
Oman	0.008887392152826168

Countries with the highest energy consumption relative to GDP (e.g., Netherlands Antilles, Trinidad and Tobago, Turkmenistan) use more energy per unit of economic output, indicating higher energy intensity in their economies.

14. What is the correlation between GDP growth and energy production growth?

Approach	Concepts Used
Aggregated Energy production Calculated Year-over-Year Changes Categorized trends Correlation Analysis	• CTE • Window functions: LAG() • Conditional Logic: CASE • Data Aggregation: SUM()


```

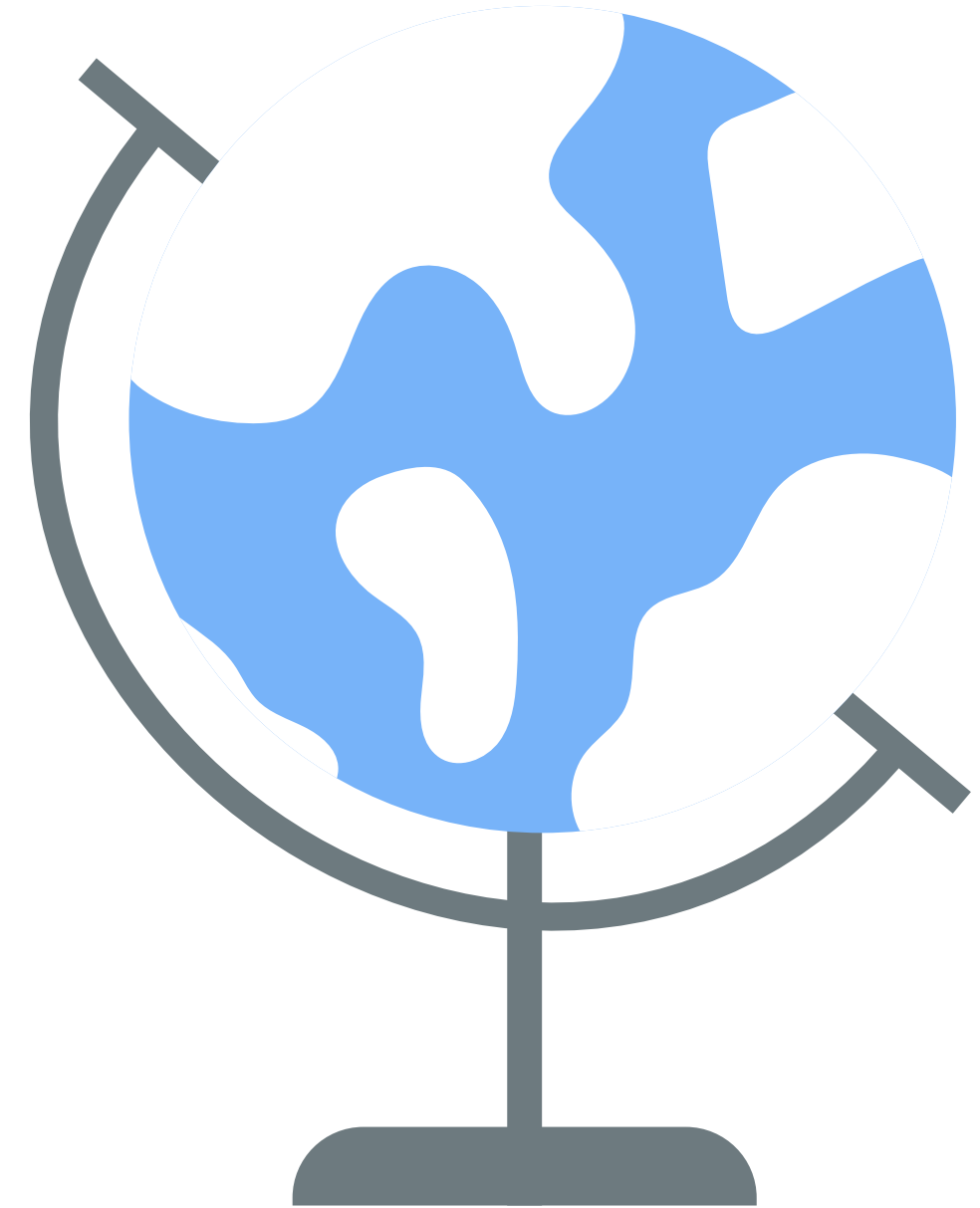
WITH TOT_PRO AS(
SELECT COUNTRY, YEAR, SUM(PRODUCTION) AS TOTAL_PRODUCTION FROM PRODUCTION GROUP BY COUNTRY, YEAR
),
TREND AS(
SELECT P.COUNTRY, P.YEAR, G.VALUE AS GDP, P.TOTAL_PRODUCTION AS PRODUCTION ,
G.VALUE - LAG(G.VALUE) OVER (PARTITION BY COUNTRY ORDER BY YEAR) AS GDP_CHANGE,
P.TOTAL_PRODUCTION - LAG(P.TOTAL_PRODUCTION) OVER (PARTITION BY COUNTRY ORDER BY YEAR) AS PROD_CHANGE
FROM TOT_PRO P
JOIN GDP_3 G ON
P.COUNTRY = G.COUNTRY AND
P.YEAR = G.YEAR
ORDER BY COUNTRY, YEAR
)
SELECT *,
CASE
WHEN GDP_CHANGE IS NULL THEN "STARTING YEAR"
WHEN GDP_CHANGE < 0 THEN "GDP Decreased"
WHEN GDP_CHANGE > 0 THEN "GDP Increased"
ELSE "NO CHANGE"
END AS GDP_TREND,
CASE
WHEN PROD_CHANGE IS NULL THEN "STARTING YEAR"
WHEN PROD_CHANGE < 0 THEN "PROD Decreased"
WHEN PROD_CHANGE > 0 THEN "PROD Increased"
ELSE "NO CHANGE"
END AS PROD_TREND
FROM TREND;

```

COUNTRY	YEAR	GDP	PRODUCTION	GDP_CHANGE	PROD_CHANGE	GDP_TREND	PROD_TREND
Afghanistan	2020	83.21645	0.0420672110	NULL	NULL	STARTING YEAR	STARTING YEAR
Afghanistan	2021	65.95827	0.0571382690	-17.258179999999996	0.0150710580	GDP Decreased	PROD Increased
Afghanistan	2022	61.84236	0.0962556700	-4.1159099999999995	0.0391174010	GDP Decreased	PROD Increased
Afghanistan	2023	58.90645	0.0180285970	-2.93591	-0.0782270730	GDP Decreased	PROD Decreased
Albania	2020	36.78752	0.0611128690	NULL	NULL	STARTING YEAR	STARTING YEAR
Albania	2021	40.04762	0.0718560580	3.2601000000000013	0.0107431890	GDP Increased	PROD Increased
Albania	2022	41.99279	0.0664429490	1.9451699999999974	-0.0054131090	GDP Increased	PROD Decreased
Albania	2023	43.63172	0.0721412460	1.6389300000000002	0.0056982970	GDP Increased	PROD Increased
Algeria	2020	531.9749	5.9694254450	NULL	NULL	STARTING YEAR	STARTING YEAR
Algeria	2021	552.5308	6.7318767720	20.555899999999995	0.7624513270	GDP Increased	PROD Increased
Algeria	2022	572.6695	6.7245775530	20.138699999999997	-0.0072992190	GDP Increased	PROD Decreased
Algeria	2023	596.0418	6.8180764570	23.372299999999996	0.0934989040	GDP Increased	PROD Increased
Angola	2020	215.9016	3.0329337790	NULL	NULL	STARTING YEAR	STARTING YEAR
Angola	2021	218.2898	2.6878279770	2.3882000000000012	-0.3451058020	GDP Increased	PROD Decreased
Angola	2022	225.5227	2.7518077110	7.2328999999999972	0.0639797340	GDP Increased	PROD Increased
Angola	2023	227.5672	2.7496385520	2.04450000000000277	-0.0021691590	GDP Increased	PROD Decreased

- A positive correlation implies that energy production growth often supports economic growth.
- A negative or weak correlation suggests other factors (like agriculture, services, or imports) influence GDP more than energy production in that country.

Global Comparisons



15. What are the top 10 countries by population and how do their emissions compare?

Approach	Concepts Used
Identified top 10 countries by population in the most recent year using a subquery and LIMIT. Joined these countries with their emission values from the latest year. Sort by emissions to compare population vs emission levels.	• WITH (CTE) for temporary result TOP_POP • JOIN to combine population and emission tables • ORDER BY and LIMIT to get top countries • Subquery to get most recent year

```

WITH LATEST_YEAR AS(
SELECT LEAST(
(SELECT MAX(YEAR) FROM POPULATION),
(SELECT MAX(YEAR) FROM EMISSION_TOTAL))
AS YEAR_TO_USE
),
TOP_COUN AS(
SELECT COUNTRIES, VALUE AS POPULATION FROM POPULATION
WHERE YEAR = (SELECT YEAR_TO_USE FROM LATEST_YEAR)
ORDER BY VALUE DESC LIMIT 10)
SELECT T.COUNTRIES,T.POPULATION, E.EMISSION FROM TOP_COUN T
JOIN EMISSION_TOTAL E
ON T.COUNTRIES = E.COUNTRY
AND E.YEAR = (SELECT YEAR_TO_USE FROM LATEST_YEAR)
ORDER BY E.EMISSION DESC;

```

COUNTRIES	POPULATION	emission
China	1422585	12195.65713
United States	336445	4795.36631
India	1438070	2820.659914
Russia	145440.5	1844.490195
Indonesia	281190.1	829.655188
Mexico	129739.7	441.0491515
Brazil	211140.7	437.7691776
Pakistan	247504.5	212.6552589
Bangladesh	171467	125.9555776
Nigeria	227882.9	114.3969139

Countries with the largest populations (China, India, USA) also have the highest emissions.

More population generally means higher total emissions.

16. Which countries have improved (reduced) their per capita emissions the most over the last decade?

Approach	Concepts Used
Calculated per-capita emissions for each country. Used window functions FIRST_VALUE and LAST_VALUE to get emissions at start and end of the decade. Computed reduction as START_PCE – END_PCE and sorted to find countries with the largest decrease.	• Window functions (FIRST_VALUE, LAST_VALUE) • Arithmetic operations • GROUP BY and ORDER BY • LIMIT


```
WITH PER_CAP AS (  
    SELECT  
        E.COUNTRY,  
        E.YEAR,  
        E.EMISSION / P.VALUE AS PER_CAPITA_EMISSION,  
        FIRST_VALUE(E.EMISSION / P.VALUE)  
            OVER (PARTITION BY E.COUNTRY ORDER BY E.YEAR) AS FIRST_YEAR_PCE,  
        LAST_VALUE(E.EMISSION / P.VALUE)  
            OVER (PARTITION BY E.COUNTRY ORDER BY E.YEAR  
                ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING) AS LAST_YEAR_PCE  
    FROM EMISSION_TOTAL E JOIN POPULATION P ON E.COUNTRY = P.COUNTRIES  
    AND E.YEAR = P.YEAR  
)  
SELECT  
    COUNTRY,  
    FIRST_YEAR_PCE AS START_PCE,  
    LAST_YEAR_PCE AS END_PCE,  
    (FIRST_YEAR_PCE - LAST_YEAR_PCE) AS IMPROVED_BY  
FROM PER_CAP  
GROUP BY COUNTRY, FIRST_YEAR_PCE, LAST_YEAR_PCE  
ORDER BY IMPROVED_BY DESC LIMIT 10;
```

Countries like Brunei and Luxembourg have achieved the largest reductions in per-capita emissions over the last decade.

COUNTRY	START_PCE	END_PCE	IMPROVED_BY
Brunei	0.0264231256761227	0.023581126312509...	0.0028419993636130333
Luxembourg	0.015099343843072...	0.013050618167430...	0.0020487256756417088
Trinidad and Tobago	0.023784443209563...	0.022375514620754...	0.0014089285888084849
United Arab Emirates	0.026841554479831...	0.025530973475110...	0.001310581004721853
Ukraine	0.004147406511321...	0.002838668113186...	0.001308738398134853
Lebanon	0.004185677969163...	0.003028288630470...	0.001157389338692825
Czechia	0.008380903217749...	0.007346339899910...	0.001034563317838461
Hong Kong	0.008840298225623...	0.007850962753740...	0.0009893354718823345
Malta	0.017232719617321...	0.016271364420924...	0.0009613551963973087
Slovenia	0.005928289743776...	0.005085011371607...	0.0008432783721686781

17. What is the global share (%) of emissions by country?

Approach	Concepts Used
Aggregated emissions per country Calculated Global Total emissions Computed Global Share(%) Sorted by contribution	• CTEs • Aggregation Functions • Cross join • Percentage Calculation • ORDER BY

```

WITH COUNTRY_EMISSION AS (
SELECT COUNTRY, SUM(EMISSION) AS TOTAL_EMISSION
FROM EMISSION_3 GROUP BY COUNTRY
),
GLOBAL_EMISSION AS(
SELECT SUM(TOTAL_EMISSION) AS GLOBAL_EMIS FROM COUNTRY_EMISSION)
SELECT C.COUNTRY, C.TOTAL_EMISSION,
(C.TOTAL_EMISSION * 100.0 / G.GLOBAL_EMIS) AS GLOBAL_SHARE
FROM COUNTRY_EMISSION C, GLOBAL_EMISSION G
ORDER BY GLOBAL_SHARE DESC;

```

China is the largest emitter, contributing nearly one-third of global emissions.

The United States is the second largest, but less than half of China's share.

India and Russia follow, together contributing about 12% of global emissions.

Smaller countries like Iran and Indonesia have relatively low shares (~2%).

COUNTRY	TOTAL_EMISSION	GLOBAL_SHARE
China	46169.0926052	32.34679022386081
United States	19226.997220999998	13.470735737881794
India	10110.665275499998	7.083690630156267
Russia	7240.2771902	5.072651729083928
Japan	4068.30575	2.8503188006548577

18. What is the global average GDP, emission, and population by year?

Approach	Concepts Used
Calculated yearly averages for GDP, population and emissions Joined Yearly Averages Order by Year	• Aggregation: AVG() • CTEs • Joins

```
WITH GDP_YEAR AS(
SELECT YEAR, AVG(VALUE) AS AVG_GDP FROM GDP_3 GROUP BY YEAR
),
POP_YEAR AS(
SELECT YEAR, AVG(VALUE) AS AVG_POP FROM POPULATION GROUP BY year
),
EMI_YEAR AS(
SELECT YEAR, AVG(EMISSION) AS AVG_EMI FROM EMISSION_TOTAL GROUP BY YEAR
)
SELECT E.YEAR,G.AVG_GDP,P.AVG_POP, E.AVG_EMI FROM EMI_YEAR E
JOIN POP_YEAR P ON E.YEAR = P.YEAR
JOIN GDP_YEAR G ON P.YEAR = G.YEAR
ORDER BY E.YEAR;
```

YEAR	AVG_GDP	AVG_POP	AVG_EMI
2020	624.400841806	39362.84443550002	152.14175302608064
2021	665.3489843160004	39690.73796750001	159.17360575268157
2022	688.6989718209995	40047.51826499997	162.46232281423315
2023	711.8175613620001	40411.240622000005	167.0233388948154

GDP Growth
Average global GDP increased steadily from 624.4 → 711.8 (2020–2023).
Indicates economic recovery or growth year over year.

Population Growth
Average population rose slightly from 39,362 → 40,411 million, showing moderate population growth.

Emission Growth
Average emissions increased from 152.14 → 167.02, roughly a 10% increase over 4 years.
Suggests economic growth is still accompanied by rising emissions.

Key Insight
Although GDP is growing, emissions are not decreasing, highlighting the need for greener energy and sustainable development policies.

FINAL BUSINESS INSIGHTS AND RECOMMENDATIONS

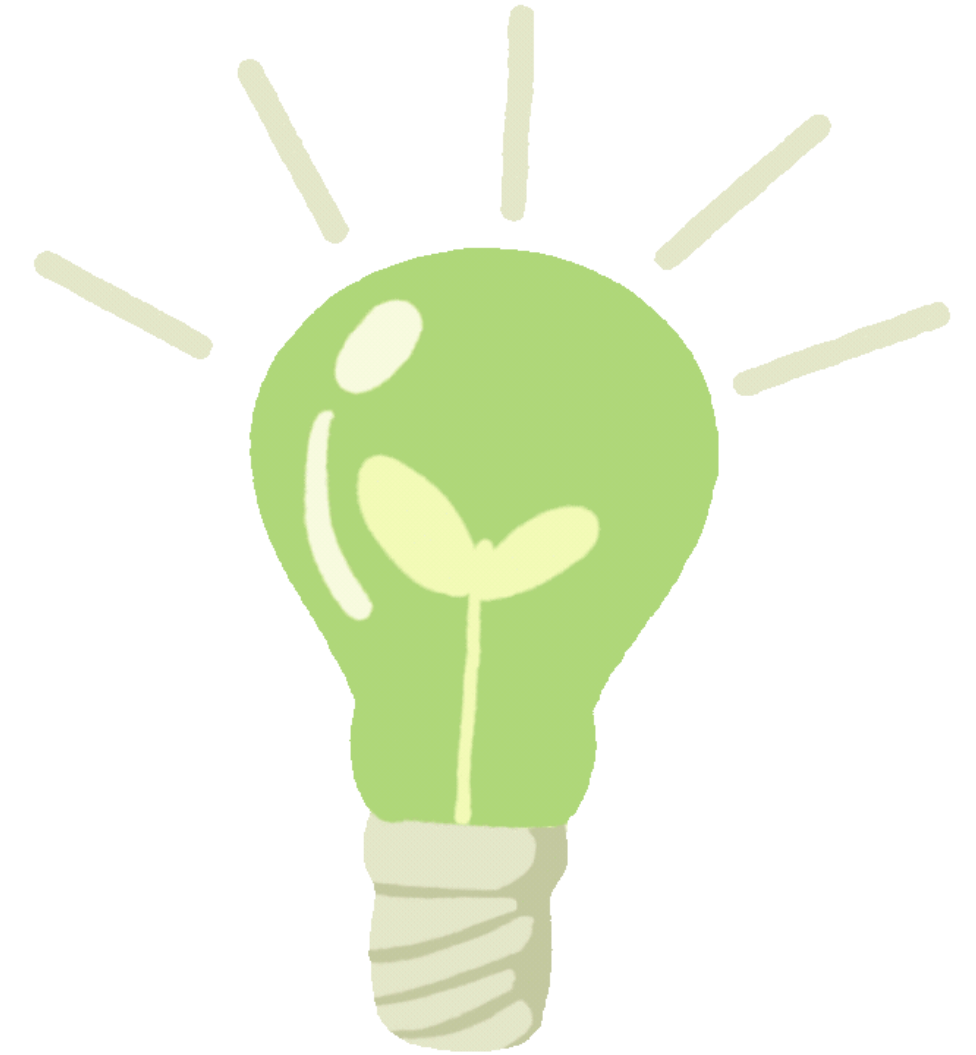
- **China** is the **top** global emitter.
- **India** has the **largest** population.
- **United states** has **high emissions** relative to population.
- **Coal** remains the **highest-emissions** energy source globally.
- **Highest per-capita** energy producers: **Qatar, Norway, Kuwait.**
- Countries **improving per-person emissions**: Brunei, Luxembourg, Trinidad and Tobago.
- Global trend: GDP, population and emissions are **rising**.
- **Populations** is **not** the **only factor** affecting the **emissions**.

FINAL BUSINESS INSIGHTS AND RECOMMENDATIONS

- 
- Focus on top emitters: China, US, India.
 - Promote cleaner energy adoption.
 - Improve energy efficiency in high-emission countries.
 - Invest in global collaboration and knowledge sharing.
 - Target coal-heavy economies.
 - Use data-driven policies

CONCLUSION

Overall, our findings emphasize the importance of targeted policies, cleaner energy adoption, and sustainable economic growth to reduce global emissions while supporting development. This SQL-driven approach provides actionable insights for governments, businesses, and policymakers to make data-driven decisions for a more sustainable future.



Experience/Challenges

During data import, null/blank values were not imported.

Approach 1 - Temp table	Approach 2 - Bulk Load
• CREATE • INSERT • VALIDATE • TRANSFER • DROP	• LOAD • NULLIFY • VALIDATE • EFFICIENT • FORMAT

Temporary table



```
CREATE TABLE POPULATION_TEMP(  
country VARCHAR(100),  
year varchar(10),  
Value varchar(50)  
);
```

```
INSERT INTO population (countries, year, Value)  
SELECT  
    country,  
    CAST(year AS UNSIGNED),  
    CASE WHEN Value = '' THEN NULL  
         ELSE CAST(Value AS DOUBLE)  
    END  
FROM POPULATION_TEMP;
```



Transfer

Experience/Challenges

During data import, null/blank values were not imported.

Approach 1 - Temp table	Approach 2 - Bulk Load
• CREATE • INSERT • VALIDATE • TRANSFER • DROP	• LOAD • NULLIFY • VALIDATE • EFFICIENT • FORMAT


```
LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\country_3.csv'  
INTO TABLE COUNTRY  
FIELDS TERMINATED BY ','  
ENCLOSED BY ''  
LINES TERMINATED BY '\\r\\n'  
IGNORE 1 LINES  
(@var_Country, @var_CID)  
SET  
    Country = NULLIF(@var_Country, ''),  
    CID = NULLIF(@var_CID, '');
```



THANK YOU

